

PH 101

Lecture 05

OUTLINE

- Energy
- Stationary Points and Minimizing Functions

Systems of Particles

The fundamental forces are those that act between particles, like gravity and electric forces. These depend on a number of things: Gravitational forces between particles are proportional to the product of their masses, and electric forces are proportional to the product of their electric charges. Charges and masses are considered to be intrinsic properties of a particle, and specifying them is part of specifying the system itself. **Apart from the intrinsic properties, the forces depend on the location of the particles.** For example, the distance between objects determines the electric and gravitational force that one particle exerts on another. Suppose that the locations of all the particles are described by their coordinates: x_1, y_1, z_1 for the **first particle**, x_2, y_2, z_2 for the **second particle**, x_3, y_3, z_3 for the **third particle**, and so on up to the last, or the Nth, particle. Then the force on any one particle is a function of its location as well as the location of all the others. We can write this in the form:

$$\vec{F}_i = \vec{F}_i \left(\left\{ \vec{r} \right\} \right).$$

What this equation means is that the force on the i -th particle is a function of the positions of all the particles.

$\left\{ \vec{r} \right\}$: stands for the collective location of every particle in the system. Another way of saying this is that the symbol represents the set of all position vectors.

Once we know the force on any particle—for example, particle number 1—we can write Newton's equation of motion for that particle:

$$\vec{F}_1 \left(\left\{ \vec{r} \right\} \right) = m_1 \vec{a}_1,$$

where m_1 and $\vec{a}_1$ are the mass and acceleration of particle 1.

When we express the acceleration as the second derivative of the position, the equation becomes

$$\vec{F}_1 \left(\left\{ \vec{r} \right\} \right) = m_1 \frac{d^2 \vec{r}_1}{d t^2}.$$

In fact, we can write such an equation for each particle:

$$\vec{F}_1 \left(\left\{ \vec{r} \right\} \right) = m_1 \frac{d^2 \vec{r}_1}{d t^2}$$

$$\vec{F}_2 \left(\left\{ \vec{r} \right\} \right) = m_2 \frac{d^2 \vec{r}_2}{d t^2}$$

$$\vec{F}_3 \left(\left\{ \vec{r} \right\} \right) = m_3 \frac{d^2 \vec{r}_3}{d t^2}$$

$\vdots$

$$\vec{F}_N \left(\left\{ \vec{r} \right\} \right) = m_N \frac{d^2 \vec{r}_N}{d t^2}$$

or, in condensed form

$$\vec{F}_i \left(\left\{ \vec{r} \right\} \right) = m_i \frac{d^2 \vec{r}_i}{d t^2}.$$

States of a System of Particles

The state of a system of particles consists of more than just their current locations; it also includes their current velocities.

For example, if the system is a single particle, its state consists of six pieces of data: the three components of its position and the three components of its velocity. We may express this by saying that the state is a point in a six-dimensional space of states labeled by axes

$$x, y, z, v_x, v_y, v_z$$

Now let's consider the motion of the particle. At each instant of time, the state is specified by the values of the six variables $x(t)$, $y(t)$, $z(t)$, $v_x(t)$, $v_y(t)$, $v_z(t)$. The history of the particle can be pictured as a trajectory through the six dimensional state-space.

Next, **consider the space of states of a system of N particles.** To specify the state of the system, we need to specify the state of every particle. This obviously means that the space of states is $6N$ -dimensional: **three position components and three velocity components for each of the N particles.** One may even say that the motion of the system is a trajectory through a $6N$ -dimensional space

$$m_i \frac{d v_i}{d t} = F_i$$

$$\frac{d r_i}{d t} = v_i.$$

Energy

It has a fundamental role in the **Lagrangian** and **Hamiltonian** formulations of mechanics.

Suppose a particle P of mass m moves under the influence of a force $\mathbf{F}$. Then its equation of motion is

$$m \frac{d\mathbf{v}}{dt} = \mathbf{F} \quad (1)$$

Here $\mathbf{v}$ is the velocity of P at time t . At this stage we place no restrictions on the force $\mathbf{F}$. It may depend on the position of P, the velocity of P, the time, or anything else; if more than one force is acting on P, then $\mathbf{F}$ means the vector resultant of these forces. On taking the scalar product of both sides of equation (1) with $\mathbf{v}$, we obtain the scalar equation

$$m\mathbf{v} \cdot \frac{d\mathbf{v}}{dt} = \mathbf{F} \cdot \mathbf{v}$$

Since,

$$m\mathbf{v} \cdot \frac{d\mathbf{v}}{dt} = \frac{d}{dt} \left(\frac{1}{2} m\mathbf{v} \cdot \mathbf{v} \right)$$

We can write

$$\frac{dT}{dt} = \mathbf{F} \cdot \mathbf{v} \quad \text{where,} \quad T = \frac{1}{2} m\mathbf{v} \cdot \mathbf{v}.$$

$$\frac{dT}{dt} = \mathbf{F} \cdot \mathbf{v} \quad (2)$$

$T = \frac{1}{2}m\mathbf{v} \cdot \mathbf{v} = \frac{1}{2}m|\mathbf{v}|^2$ is called the kinetic energy of the particle P.

If we now integrate equation (2) over the time interval $[t_1, t_2]$, we obtain

$$T_2 - T_1 = \int_{t_1}^{t_2} \mathbf{F} \cdot \mathbf{v} dt$$

where T_1 and T_2 are the kinetic energies of P at times t_1 and t_2 respectively. This is the **energy principle** for a particle moving under a force $\mathbf{F}$.

$W = \int_{t_1}^{t_2} \mathbf{F} \cdot \mathbf{v} dt$ is called the work done by the force $\mathbf{F}$ during the time interval $[t_1, t_2]$. The rate of working of $\mathbf{F}$ at time t is thus $\mathbf{F} \cdot \mathbf{v}$

Energy principle for a particle

In any motion of a particle, the increase in the kinetic energy of the particle in a given time interval is equal to the total work done by the applied forces during this time interval.

Suppose that the particle P moves along the x-axis under the force F acting in the positive x-direction. In this case, the ‘work done’ integral

$$W = \int_{t_1}^{t_2} \mathbf{F} \cdot \mathbf{v} dt \quad \text{reduces to:}$$

$$W = \int_{x_1}^{x_2} F(x) dx \quad \text{Here } x_1 = x(t_1) \text{ and } x_2 = x(t_2)$$

(This is a common definition of the work done by a force F . It can be used when $F = F(x)$, but not in general.) It follows that the energy principle for a particle moving in a rectilinear force field can be written

$$T_2 - T_1 = \int_{x_1}^{x_2} F(x) dx.$$

Now let $V(x)$ be the indefinite integral of $-F(x)$, so that

$$F = -\frac{dV}{dx}$$

$$\int_{x_1}^{x_2} F(x) dx = V(x_1) - V(x_2)$$

Such a V is called the potential energy function of the force field F . In terms of V , the energy principle in rectilinear motion can be written

$$T_2 + V(x_2) = T_1 + V(x_1)$$

Or,
$$T + V = E$$

where E is a constant called the total energy of the particle.

GENERAL FEATURES OF RECTILINEAR MOTION

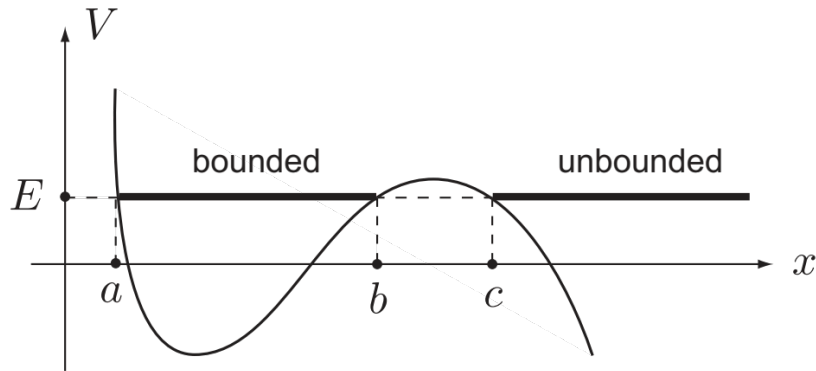
$$\frac{1}{2}mv^2 + V(x) = E$$

Since $T \geq 0$ (and is equal to zero only when $v = 0$) it follows that the position of the particle is restricted to those values of x that satisfy

$$V(x) \leq E$$

The equality will occur only when $v = 0$.

Suppose that $V(x)$ has the form shown in Figure



The motion of the particle must take place either (i) in the bounded interval $a \leq x \leq b$, or (ii) in the unbounded interval $c \leq x \leq \infty$. Thus, if the particle was situated in the interval $[a, b]$ initially, this is the interval in which the motion will take place.

Bounded motion

$$\frac{1}{2}mv^2 + V(x) = E$$

Suppose that the motion is started with P in the interval $[a, b]$ and with v positive, so that P is moving to the right. Then, since v can only be zero at $x = a$ and $x = b$, v will remain positive until P reaches the point $x = b$, where it comes to rest. From equation , it follows that the ODE that governs this ‘right’ part of the motion is

$$\frac{dx}{dt} = + \left[\frac{2(E - V(x))}{m} \right]^{1/2}$$

At the point $x = b$, $dV/dx > 0$ which implies that $F < 0$. P therefore moves to the left and does not stop until it reaches the point $x = a$. The ODE that governs this ‘left’ part of the motion is

$$\frac{dx}{dt} = - \left[\frac{2(E - V(x))}{m} \right]^{1/2}$$

At the point $x = a$, $dV/dx < 0$, which implies that $F > 0$ and that P moves to the right once again. The result is that P performs **periodic oscillations** between the extreme points $x = a$ and $x = b$.

Since the 'left' and 'right' parts of the motion take equal times, the period τ of these oscillations can be found by integrating either equation over the interval $a \leq x \leq b$. Each equation is a separable ODE and integration gives

$$\tau = 2 \int_a^b \frac{dx}{\underbrace{[2(E - V(x))]}_m^{1/2}}$$

It should be noted that these oscillations are generally not simple harmonic. In particular, their period is *amplitude dependent*.

Unbounded motions

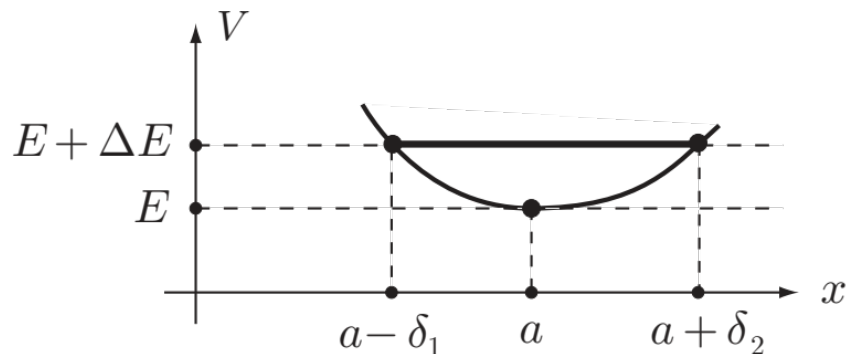
Suppose now that the motion is started with P in the interval $[c, \infty)$ and with v negative, so that P is moving to the left. Then, since v can only be zero at $x = c$, v will remain negative until P reaches the point $x = c$, where it comes to rest. At the point $x = c$, $dV/dx < 0$ which implies that $F > 0$. P therefore moves to the right and continues to do so indefinitely

Equilibrium positions of a particle

The point A is said to be an **equilibrium position** of a particle P if, when P is released from rest at A, P remains at A

In the case of rectilinear motion under a force field $F(x)$, the point $\mathbf{x} = \mathbf{a}$ will be an equilibrium position of P if (and only if) $\mathbf{F}(\mathbf{a}) = 0$, that is, if dV/dx (at $x=a$) = 0. The equilibrium positions of P are the **stationary points of the potential energy function $V(x)$** .

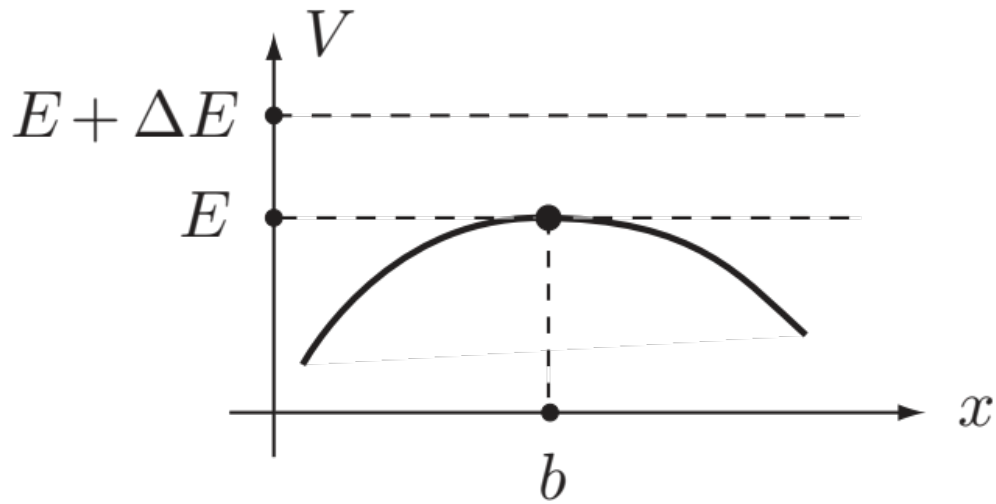
Stable Equilibrium



$$\left. \frac{dV}{dx} \right|_{x=a} = 0$$

$$\left. \frac{d^2V}{dx^2} \right|_{x=a} > 0$$

Unstable Equilibrium



$$\left. \frac{dV}{dx} \right|_{x=b} = 0$$

$$\left. \frac{d^2V}{dx^2} \right|_{x=b} < 0$$

Approximate equation of motion for small oscillations

Suppose that the point $x = a$ is a minimum point of the potential energy $V(x)$. Then, when x is sufficiently close to a , we may approximate $V(x)$ by the first three terms of its Taylor series in powers of the variable $(x - a)$, as follows:

$$\begin{aligned} V(x) &= V(a) + (x - a)V'(a) + \frac{1}{2}(x - a)^2V''(a) \\ &= V(a) + \frac{1}{2}(x - a)^2V''(a), \end{aligned}$$

$$\left. \frac{dV}{dx} \right|_{x=a} = V'(a) = 0$$

Here,

$$\left. \frac{d^2V}{dx^2} \right|_{x=a} = V''(a)$$

Thus, for small amplitude oscillations about $x = a$, the energy conservation equation is approximately:

$$\frac{1}{2}mv^2 + V(a) + \frac{1}{2}(x - a)^2 V''(a) = E$$

If we now differentiate this equation with respect to t (and divide by v), we obtain the approximate (linearised) equation of motion

$$m \frac{d^2 x}{dt^2} + V''(a) (x - a) = 0$$

Provided that $V''(a) > 0$, this is the equation for **simple harmonic oscillations** with angular frequency $(V''(a)/m)^{1/2}$ about the point $x = a$. The small oscillations of P about $x = a$ are therefore **approximately simple harmonic** with **approximate period** $\tau = 2\pi(m/V''(a))^{1/2}$.

Recall the Gradient operator!

If $\psi(\mathbf{r})$ is a scalar field then $\text{grad } \psi$ is the vector field defined by

$$\vec{\nabla} \psi = \text{grad } \psi = \frac{\partial \psi}{\partial x} \mathbf{i} + \frac{\partial \psi}{\partial y} \mathbf{j} + \frac{\partial \psi}{\partial z} \mathbf{k}$$

Conservative field

If the field $\mathbf{F}(\mathbf{r})$ can be expressed in the form

$$\mathbf{F} = -\text{grad } V = -\vec{\nabla} V$$

where $V(\mathbf{r})$ is a scalar function of position, then $\mathbf{F}$ is said to be a **conservative field** and the function V is said to be the potential energy function for $\mathbf{F}$

Example:

Show that the field $\mathbf{F}_1 = -2x\mathbf{i} - 2y\mathbf{j} - 2z\mathbf{k}$ is conservative but that the field $\mathbf{F}_2 = y\mathbf{i} - x\mathbf{j}$ is not.

Solution:

(i) If $\mathbf{F}_1$ is conservative, then its potential energy V must satisfy

$$-\frac{\partial V}{\partial x} = -2x, \quad -\frac{\partial V}{\partial y} = -2y, \quad -\frac{\partial V}{\partial z} = -2z,$$

and these equations integrate to give

$$V = x^2 + p(y, z), \quad V = y^2 + q(x, z), \quad V = z^2 + r(x, y),$$

where p , q and r are ‘constants’ of integration, which, in this case, are functions of the other variables. If V really exists, then these three representations of V can be made identical by making special choices of the functions p , q and r . In this example it is clear that this can be achieved by taking $p = y^2 + z^2$, $q = x^2 + z^2$, and $r = x^2 + y^2$. Hence $\mathbf{F}_1 = -\text{grad}(x^2 + y^2 + z^2)$ and so $\mathbf{F}_1$ is conservative.

(ii) If $\mathbf{F}_2$ is conservative, then its potential energy V must satisfy

$$-\frac{\partial V}{\partial x} = y, \quad -\frac{\partial V}{\partial y} = -x, \quad -\frac{\partial V}{\partial z} = 0.$$

There is no V that satisfies these equations simultaneously. The easiest way to show this is to observe that, from the first equation, $\partial^2 V / \partial y \partial x = -1$ while, from the second equation, $\partial^2 V / \partial x \partial y = +1$. Since these mixed partial derivatives of V should be equal, we have a contradiction. The conclusion is that no such V exists and that $\mathbf{F}_2$ is not conservative. ■

How to check quickly if a given force field, $\vec{F}$, is conservative?

A given force $\vec{F}$ is conservative if the cross-product of the $\vec{\nabla}$ -operator and $\vec{F}$ is zero.

$$\vec{\nabla} \times \vec{F} = 0$$

This effectively means, we can write $\vec{F}$ as the gradient of a scalar function, say V , termed as the potential energy.

$$\vec{F} = -\vec{\nabla} V$$

Now, recall in cartesian coordinates

$$\begin{aligned}\vec{A} \times \vec{B} &= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} \\ &= \hat{x} (A_y B_z - A_z B_y) + \hat{y} (A_z B_x - A_x B_z) \\ &\quad + \hat{z} (A_x B_y - A_y B_x)\end{aligned}$$

In the similar way, in cartesian
coordinates we can write:

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$

$$= \hat{x} \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) + \hat{y} \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) \\ + \hat{z} \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right)$$

If $\vec{F}$ is conservative $\vec{\nabla} \times \vec{F} = 0$

$\Rightarrow$ we must have:

$$\frac{\partial F_z}{\partial y} = \frac{\partial F_y}{\partial z} ; \quad \frac{\partial F_x}{\partial z} = \frac{\partial F_z}{\partial x} \quad \text{and}$$

$$\frac{\partial F_y}{\partial x} = \frac{\partial F_x}{\partial y}$$

check if $\vec{F} = -2x\hat{i} - 2y\hat{j} - 2z\hat{k}$
is conservative.

Solution

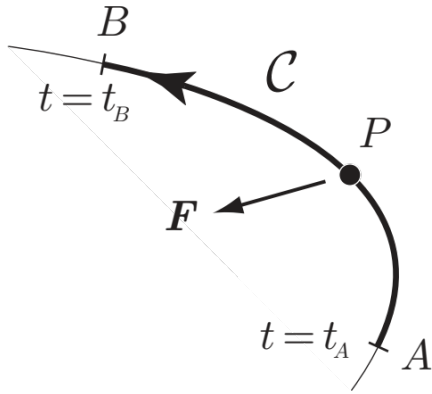
$$F_x = -2x, \quad F_y = -2y, \quad F_z = -2z$$

$$\begin{aligned}\vec{\nabla} \times \vec{F} &= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -2x & -2y & -2z \end{vmatrix} \\ &= \hat{x}(0-0) + \hat{y}(0-0) \\ &\quad + \hat{z}(0-0) \\ &= 0\end{aligned}$$

Thus, $\vec{F}$ is conservative!

ENERGY CONSERVATION IN A CONSERVATIVE FIELD

Suppose now that the particle P is in general three-dimensional motion under the force $\mathbf{F}$ and that, in the time interval $[t_A, t_B]$, P moves from the point A to the point B along the path C, as shown in Figure. Then, by the energy principle,



$$T_B - T_A = \int_{t_A}^{t_B} \mathbf{F} \cdot \mathbf{v} dt \quad (1)$$

where T_A and T_B are the kinetic energies of P when $t = t_A$ and $t = t_B$ respectively. When $\mathbf{F}$ is a force field $\mathbf{F}(\mathbf{r})$, the ‘work done’ integral on the right side of equation (1) can be written in the form

$$\int_{t_A}^{t_B} \mathbf{F} \cdot \mathbf{v} dt = \int_{t_A}^{t_B} \mathbf{F}(\mathbf{r}) \cdot \frac{d\mathbf{r}}{dt} dt = \int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} \quad (2)$$

Thus,

$$T_B - T_A = \int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} \quad (3)$$

Suppose now that the field $\mathbf{F}(\mathbf{r})$ is conservative with potential energy $V(\mathbf{r})$ and let C be any path connecting the points A and B . Then

$$\begin{aligned} W[A \rightarrow B; C] &= \int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} = \int_C (-\text{grad } V) \cdot d\mathbf{r} = \int_C (-\vec{\nabla} V) \cdot d\vec{\mathbf{r}} \\ &= - \int_C \left(\frac{\partial V}{\partial x} \mathbf{i} + \frac{\partial V}{\partial y} \mathbf{j} + \frac{\partial V}{\partial z} \mathbf{k} \right) \cdot (dx \mathbf{i} + dy \mathbf{j} + dz \mathbf{k}) \\ &= - \int_C \frac{\partial V}{\partial x} dx + \frac{\partial V}{\partial y} dy + \frac{\partial V}{\partial z} dz = - \int_C dV = V_A - V_B. \end{aligned}$$

Thus, when $\mathbf{F}$ is conservative with potential energy V ,
$$\int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} = V_A - V_B \quad (4)$$

for any path C connecting the points A and B . The energy principle can therefore be written, from (3) and (4)

$$T_B + V_B = T_A + V_A$$

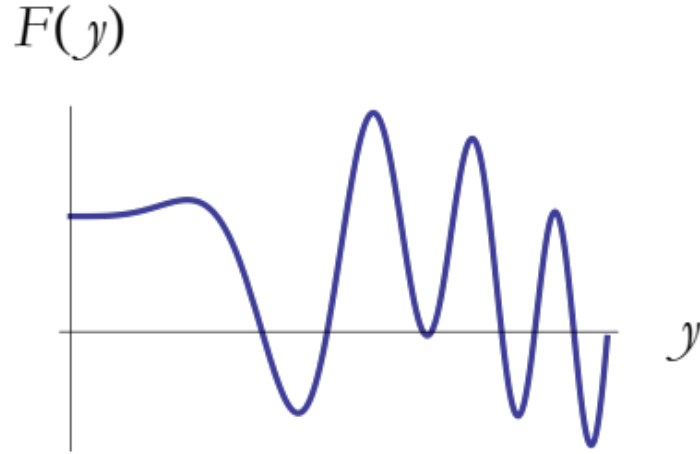
which is equivalent to the energy conservation formula
$$T + V = E$$

Energy conservation in 3-D motion

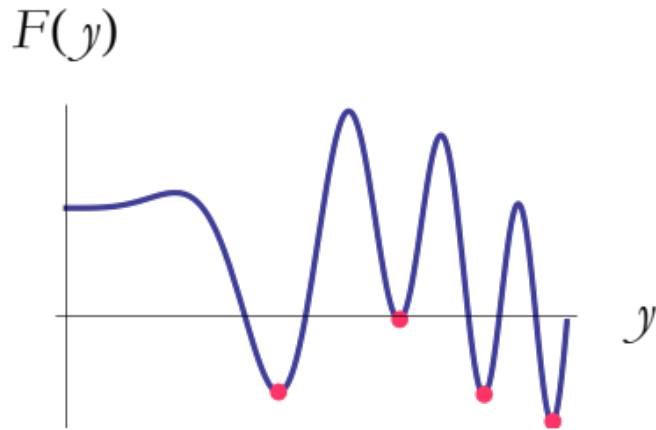
When a particle moves in a **conservative** force field, the sum of its kinetic and potential energies remains constant in the motion.

Stationary Points and Minimizing Functions

Let's look at a function of y that we call, say **F**



Notice that there are places on the curve where a shift in y in either direction produces only an upward shift in F . These points are called *local minima*.



Local minima -> (look at the red dots)

For each local minimum, when you go in either direction along y , you begin to rise above the dot in $F(y)$. Each dot is at the bottom of a little depression. The global minimum is the lowest possible place on the curve.

One condition for a local minimum is that the derivative of the function with respect to the independent variable at that point is zero. This is a necessary condition, but not a sufficient condition. This condition defines any **stationary point**:

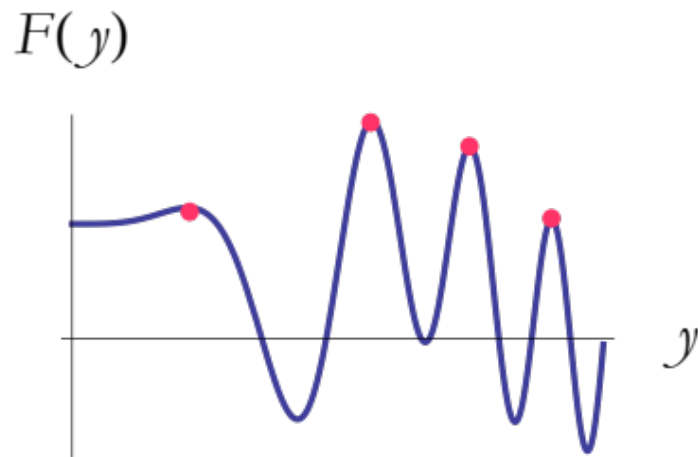
$$\frac{d}{dy}F(y) = 0$$

The second condition tests to see what the character of the stationary point is by examining its second derivative. If the second derivative is larger than 0, then all points nearby will be above the stationary point, and we have a **local minimum**:

$$\frac{d^2}{dy^2}F(y) > 0$$

If the second derivative is less than 0, then all points nearby will be below the stationary point, and we have a **local maximum**:

$$\frac{d^2}{dy^2} F(y) < 0$$



If the second derivative is equal to 0, then the derivative changes from positive to negative at the stationary point, which we call a **point of inflection**:

$$\frac{d^2}{dy^2} F(y) = 0$$

